PlayProbe

Customization, Functions, Use Cases and Test Report

Project	PlayProbe (folder: simple-abr-player)
Base	Open-source PlayProbe by garghe, MIT License
Report date	29 October 2026
Scope	Frontend redesign and monitoring features, one backend change, test cases and use cases
Prepared for	Project owner and reviewers

> **Status:** every change type-checks and builds (`npm run build`), and the 157 existing automated tests pass. The new screens have **not** been verified in a recorded browser run and no new automated tests were written; Section 8 lists manual test cases for that.

1. The open-source base

Item	Detail
Name	PlayProbe (repository folder `simple-abr-player`)
Author / license	Copyright 2026 garghe. **MIT License**: free to use, modify and redistribute provided the copyright and license notice stay in the project.
Purpose	A web app for stream engineers. It plays MPEG-DASH and HLS streams, shows a "stats for nerds" analysis panel, and monitors SRT, UDP and HLS feeds on a live multiviewer wall with alerts.
Frontend stack	Plain TypeScript (no React or Vue), a small DOM helper (`src/ui/dom.ts`), one hand-written CSS file, Vite 8 build, Shaka Player 5 for playback, canvas and SVG for charts, Web Audio API for loudness.
Backend stack	Node server (`server/`) that runs ffmpeg and srt-live-transmit sessions, analyses MPEG-TS, evaluates alerts and sends notifications. A Cloudflare Worker (`worker/`) is the static-hosting option.
Testing	Vitest, 14 test files, 157 tests.

What the base project already provided (not part of our work)

- **Player:** controls, seek bar, quality / audio / subtitle selection, picture-in-picture, fullscreen, custom headers, CORS proxy.
- **Stats tabs:** Overview, Tracks, Audio (EBU R128 / ATSC A/85 loudness and meters), SCTE-35, Network, Manifest, DRM, Device, Log, Alerts, SRT.
- **DRM:** Widevine and FairPlay with any license server.
- **SCTE-35:** marker decoding, ad-break detection, manual decoder.
- **SRT / UDP / HLS monitoring:** channel sessions, link statistics, MPEG-TS health (continuity, PCR, PAT/PMT), black video and silence detection, the alert engine, Slack / Teams / webhook / email notifications and daily log files.

2. Summary of what we added

ID	Layer	Feature
F1	Frontend	New layout from your sketch: top bar, left sidebar, centre monitoring tiles, right column with Full preview above Full analysis
F2	Frontend	Preview and analysis column opens only when a tile, URL or test stream is chosen
F3	Frontend	Close (x) button on Full preview so tiles get the full width
F4	Frontend	Collapsible sidebar (three-line Menu button) holding Test streams, Load, SRT and DRM
F5	Frontend	Animated circular CPU gauge with limit marker, replacing the CPU bar and the old "Add channels" bar position
F6	Frontend	Add channel menu with two options: Add by URL list, Add manually

F7	Frontend	Grid sizes Auto, 2x2, 3x3, 4x4, 5x5, 6x6, 8x8 instead of Small / Medium / Large
F8	Frontend	Inner scroll bar removed; the page scrolls instead
F9	Frontend	Fullscreen tele-wall: tiles fill the whole screen with no empty space for any channel count, with a playing / alerts badge
F10	Frontend	Alert popups (toasts) removed
F11	Frontend	Tiles show only Black video, Audio silence and Ad break; SCTE IN / other cue badges and link alerts removed from tiles
F12	Frontend	History chart with alert markers, pan back in time, time-window selector, date and time "Go to", Live button, zoom, tooltips
F13	Frontend	Clickable legend: click Packet loss, Recovered, Black video and so on to see only that
F14	Frontend	History stored in the browser (IndexedDB) for 7 days so it survives a reload
F15	Frontend	Human-readable alert names (for example link_loss shown as "Packet loss")
F16	Frontend	Bug fix: the CPU ring's limit marker was drawn 90 degrees off and is now placed correctly
B1	Backend	Stricter black-video detection: only a fully black picture raises the alert

3. Frontend changes in detail

3.1 New page layout (F1, F2, F3, F4)

> *Your request: use the hand-drawn UI sketch. Preview and analysis only when a tile is clicked, with a close button. Put the left menu behind a menu button.*

- **Top bar:** logo and name, the URL box, the theme button (shows "System" by default) and Settings.
- **Left sidebar:** a Menu button. When open it shows the Test streams dropdown, Load, SRT and DRM. When closed it shrinks to one button so the tiles get more room.
- **Centre:** the monitoring tiles (the SRT panel and wall).
- **Right column:** Full preview (the player) above Full analysis (the stats tabs). It stays hidden until you click a tile, press Load or choose a test stream. The x button pauses the video and hides it. The drag handle still resizes it; the default width is 560 px.
- **Small screens (under 1100 px):** everything stacks into one column.
- All element IDs were kept, so the player, stats and DRM logic were not touched.

3.2 CPU usage (F5, F16)

> *Your request: show CPU usage in a circular, more advanced animated way, in place of the "Add channels" bar.*

- SVG ring with a gradient arc that fills smoothly and glows. Green normally, amber near the limit, red when the limit is reached.
- The percentage sits in the centre. Beside it: cores text, the "No new channel above __ %" input, headroom note and the machine / server load line.
- A white marker shows the limit on the ring. It was first drawn about 90 degrees off; this was corrected so 70 percent sits at about 8 o'clock and the arc stops short of it while usage is below the limit.
- The glow animation is switched off for users who prefer reduced motion.

3.3 Adding channels (F6)

> *Your request: a button that offers two ways to add a channel, by URL or manually, and opens only the chosen one.*

- Button **Add channel** opens a small menu: **Add by URL list** (paste one or many lines, optional names, Start-now option) or **Add manually** (name and address, Start / Stop / Delete).
- Only the chosen form is visible; choosing the same option again closes it. Clicking a tile, or typing an srt:// or udp:// address in the main URL box, opens the manual form for that channel.
- The collapsed "Add channels" header bar and the "Logs: ... path" line at the bottom were removed.

3.4 Monitoring wall (F7, F8, F9)

> *Your request: grid sizes like 3x3 or 4x4, one outer scroll bar only, and a fullscreen tele-wall with no black space.*

- **Grid selector:** Auto, 2x2, 3x3, 4x4, 5x5, 6x6, 8x8 (your old saved Small / Medium choice falls back to Auto).
- **Scrolling:** the tile area no longer scrolls inside itself; the page scrolls. Tile picture loading now follows the page scroll.
- **Fullscreen:** the tiles use the whole screen whatever the number of channels. Auto chooses the number of columns whose cells are closest to 16:9 for the screen. A chosen grid uses that many columns. The channels are shared evenly over the rows and a row with fewer channels gets wider tiles, so there are no empty cells (for example, 7 channels become 4 above and 3 wider below). Pictures fill their cell, with a little cropping instead of black bars. A badge shows "N playing, N shown, N alerts". Filters (for example Problems only) are respected.

3.5 Alerts on tiles and popups (F10, F11)

> *Your request: no alert popups; tiles should show only black video, audio silence and ad breaks, not packet loss or SCTE cues.*

- Alerts no longer appear as popups. Sound and desktop notifications still follow the settings in the Alerts tab.
- Tiles show only **BLACK VIDEO** and **AUDIO SILENCE** (worst first, "+N" if more, hover for the list), plus the **AD BREAK** badge with its countdown. SCTE IN and other cue badges, and link alerts such as packet loss and RTT, are not shown on tiles. They remain in the Alerts tab and on the chart.
- The tile border colour still follows every alert, including link alerts.

3.6 History chart (F12, F13, F14, F15)

> *Your request: store alerts (SRT transmit, packet loss, recovered) and show them on the bitrate graph; make the graph movable to look back, add a date and time selector, make the graph look more defined, and make the legend names clickable to filter.*

- **Markers:** each alert is drawn at the time it was raised (colour by severity: red critical, amber warning, blue info). A green dashed marker shows when it recovered. A tinted band spans the time it was active. One-off events are purple. Hovering a marker shows the alert name, time and message. Keep-alive splice_null messages get no marker.
- **Moving through time:** drag the chart, use the back / forward buttons, pick a window (1 min, 3 min, 15 min, 1 hour, 6 hours, 24 hours, 7 days), use **Go to** to centre on a chosen date and time, press **Live** to return to now, Ctrl + mouse wheel to zoom the window.
- **Look:** taller chart, labelled axes (bitrate left, RTT right), gradient fill under the bitrate line, smooth lines, breaks where no data was recorded.
- **Legend filter:** buttons for Bitrate, RTT, each alert type seen (named, for example Packet loss, High RTT, Black video), Recovered and Events, each with a count. Click one to show only that; click again or "Show all" to reset.
- **Storage:** bitrate / RTT samples (every 5 seconds) and alerts are saved in the browser's IndexedDB for 7 days, so the chart can look back past a page reload. It records only while PlayProbe is open in that browser.

4. Backend change in detail (B1)

> *Your request: raise the black video alert only when the full screen is black, not when part of the video is black.*

- **Where:** `server/srt/session.ts`, the ffmpeg `blackframe` filter.
- **What changed:** the brightness cutoff was lowered from 32 to 20 (true black is about 16). A frame is black only when at least 98 percent of it is near-black, so dark scenes and partly black pictures no longer count. A logo or clock on an otherwise black picture still counts as black.
- **To apply:** restart the PlayProbe server and any running channels.
- This is the only backend change. The alert engine, thresholds, notifications and TS analysis were not modified.

5. Use cases

Use case	How it works now
Operations wall (NOC)	Choose a grid (for example 4x4), press Fullscreen wall. Every channel fills the screen with no empty space; the badge shows how many are playing and how many alerts are active.
Detect a dead picture or sound	A tile shows BLACK VIDEO or AUDIO SILENCE; ad breaks show AD BREAK with a countdown.
Inspect one channel	Click its tile: Full preview and Full analysis open. Close with x to give the tiles the room back.
Investigate an incident later	Select the channel, open the SRT tab, drag back or use Go to, hover the red and green markers to see what happened and when it recovered.
Isolate one problem type	Click Packet loss, Recovered, Black video and so on in the legend to see only those.
Add many channels at once	Add channel, Add by URL list, paste the lines, optionally start them one by one.
Add or edit one channel	Add channel, Add manually (name and address), or click a tile to edit it.
Check capacity before adding	The CPU ring shows load against the limit; new channels will not start above it.
Play and analyse a single stream	Paste a URL or use Menu, Test streams, Load; preview and analysis open on the right.
Avoid false black-video alerts	Dark scenes and partly black pictures no longer raise the alert.

6. Automated tests (existing)

Run with `npm test` (Vitest). Result after our changes: **14 files, 157 tests, all passing**. `npm run build` (TypeScript check plus production build) also passes.

Test file	Covers
drm.test.ts	DRM settings, PSSH and license handling
hls-cues.test.ts, srt-hls-cues.test.ts	Ad / SCTE cue tags in HLS playlists
loudness.test.ts	EBU R128 / ATSC A/85 loudness maths
mp4.test.ts	MP4 parsing for audio analysis
proxy.test.ts, worker.test.ts	CORS proxy and Cloudflare Worker
scte35.test.ts	SCTE-35 decoding
srt-alerts.test.ts	Alert rules, raise and clear behaviour
srt-channels.test.ts	Channel management
srt-hls.test.ts	HLS channel monitoring

srt-ts.test.ts	MPEG-TS analysis
srt-url.test.ts	srt:// / udp:// / https address parsing
srt-wall-scte.test.ts	The SCTE ad-break badge logic used by the wall

Not covered: we did not add automated tests for the new layout, grid, fullscreen fit, chart, legend filter or history storage. The manual cases below cover them.

7. How to run

- `npm run dev` starts the app with the local server (needed for SRT, UDP and HLS monitoring).
- `npm run build` then `npm run preview` runs the production build.
- After the backend change, restart the server and the running channels.

8. Manual test cases

ID	Area	Steps	Expected result	Result
T1	Layout	Open the app with nothing selected.	Top bar, collapsed sidebar and tiles; no preview column.	
T2	Preview opens	Click a tile.	Full preview and Full analysis appear on the right; tiles narrow.	
T3	Preview from URL	Paste a stream URL, press Enter.	Preview column opens and the stream loads.	
T4	Close preview	Click x on Full preview.	Video pauses, column hides, tiles use the full width; clicking a tile reopens it.	
T5	Sidebar	Click Menu twice.	Opens (Test streams, Load, SRT, DRM), then closes to one button; tiles widen.	
T6	Test stream	Menu, choose a test stream, Load.	Stream loads in the preview.	
T7	CPU ring	Watch the ring while channels run; change the limit.	Arc matches the percentage; marker is at the limit (70 percent about 8 o'clock); amber near the limit, red above; marker moves with the limit.	
T8	Add by URL	Add channel, Add by URL list, paste lines, Add channels.	Only the URL form shows; channels appear as tiles; choosing it again hides the form.	
T9	Add manually	Add channel, Add manually.	Only the name / address form shows.	
T10	Grid sizes	Switch Auto, 2x2, 3x3, 4x4, and so on.	Column count changes; no inner scroll bar; page scrolls.	
T11	Fullscreen, auto	Fullscreen wall with 7 channels.	All 7 fill the screen, no black area (for example 4 above, 3 wider below).	
T12	Fullscreen, fixed	Choose 3x3 with 5 channels, fullscreen.	3 columns, 2 rows, rows filled edge to edge; badge text is correct.	
T13	Fullscreen, filter	Filter to Problems only, fullscreen.	Only the shown tiles fill the screen.	

T14	No popups	Trigger an alert (stop a source, or wait for an SCTE message).	No toast appears; the alert is in the Alerts tab and on the chart.	
T15	Tile names	Feed black video, then silence, then an ad break.	BLACK VIDEO, AUDIO SILENCE, AD BREAK shown. Packet loss and SCTE IN / splice_null show nothing on the tile.	
T16	Chart live	Select a running channel, open the SRT tab.	Bitrate (and RTT with link stats) draw live; the legend shows the latest values.	
T17	Chart markers	After an alert has raised and recovered.	Coloured marker at raise, green dashed marker at recovery, tinted band between; hover shows name, time, message.	
T18	Pan back	Drag the chart to the right, then press Live.	Time moves backward with "(not live)" status; Live returns to now.	
T19	Go to	Pick a past date and time.	Window centres on it; "No data recorded for this time" if there is none.	
T20	Window	Choose 1 hour, 24 hours, 7 days.	Axis labels change and include dates when spanning days.	
T21	Legend filter	Click Packet loss, then click again; repeat with Recovered and Black video.	Only those markers show (lines fade), then everything returns. Show all also resets.	
T22	Persistence	Let a channel run a few minutes, reload, drag back.	Earlier data and alerts are still on the chart.	
T23	Black, partial	Play a source with a black area or a dark scene (server restarted).	No Black video alert.	
T24	Black, full	Play a fully black picture for longer than the black threshold.	Black video alert raises, and recovers when the picture returns.	
T25	Responsive	Narrow the window below 1100 px.	One column; sidebar becomes a row; preview stacks under the tiles.	
T26	Themes	Click the System button repeatedly.	Cycles System, Light, Dark; chart and ring stay readable.	

9. Known limits and recommendations

- **History is stored in the browser.** It records only while PlayProbe is open, is not shared between browsers or computers, is lost if site data is cleared, and is kept 7 days at 5-second resolution. For always-on history a server-side store would be needed (a separate backend change).
- **Black video:** if a truly black picture with a large logo is not detected, lower `amount` or raise `threshold` in `server/srt/session.ts`. For a strict "every pixel black" rule, raise `amount` to 100 (may miss black with compression noise).

- **No UDP sidebar button:** SRT, UDP and HLS channels are all managed from the same tiles and the Add channel menu.
- **Tile border colour** still reflects all alerts, not just the three shown by name.
- **Recommended next steps:** run the manual cases above, then add automated tests for the grid / fullscreen fit calculation and the chart legend filter.

10. Files changed

File	Change
index.html	New layout structure, sidebar, close button
src/main.ts	Show / hide preview column, sidebar toggle, no alert popups, panel opens on start
src/styles.css	Layout, sidebar, CPU ring, grid, fullscreen wall, chart toolbar and legend, tile badges
src/ui/splitter.ts	Default right column width 560 px
src/ui/srt/panel.ts	CPU ring, Add channel menu, grid selector, removed logs line
src/ui/srt/wall.ts	Grid sizes, fullscreen fit, tile badges and alert names
src/ui/srt/chart.ts	Rewritten: history chart with markers, panning, legend filter
src/ui/stats/srt.ts	Uses the new chart
src/srt/client.ts	Feeds samples and alerts to the history store
src/srt/history-store.ts *(new)*	IndexedDB history storage
src/srt/rule-labels.ts *(new)*	Readable alert names
server/srt/session.ts	Stricter black-frame detection (backend)